

Renato Velozo Ruiz

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EMPLOYMENT

- 2026– **Visiting Researcher** in Departamento de Ingeniería Matemática, Universidad de Chile.
- 2025–2026 **Research Associate** in Department of Mathematics, Imperial College London (UK).
- 2023–2025 **Postdoctoral Fellow** in Department of Mathematics, University of Toronto (Canada).
- 2022–2023 **Postdoctoral Fellow** in Laboratoire Jacques-Louis Lions, Sorbonne Université (France).

EDUCATION

- 2018–2023 **PhD in Mathematics** at University of Cambridge (UK).
- Supervisors: Mihalis Dafermos and Clément Mouhot.
- Thesis: **Linear and non-linear collisionless many-particle systems.**
- 2017–2018 **Master of Science in Mathematics** at Universidad Católica de Chile.
- 2013–2016 **Bachelor in Mathematics** at Universidad Católica de Chile.

RESEARCH INTERESTS

Analysis of PDEs. Mathematical Physics. General Relativity. Kinetic Theory.

SCIENTIFIC WORK

- Future global stability of Maxwell–Jüttner equilibria and vacuum for the massless Boltzmann equation on FLRW spacetimes, with Robert M. Strain and Martin Taylor; arXiv:2606.00175 preprint, 71 pages.
- Phase mixing and the Vlasov equation in cosmology, with Martin Taylor; arXiv:2512.04214 preprint, 52 pages.
- Decay properties for massive Vlasov fields on Schwarzschild spacetimes; arXiv:2411.05124 preprint, 67 pages.
- Decay properties of Vlasov fields on non-trapping asymptotically hyperbolic manifolds, with Anibal Velozo Ruiz; arXiv:2312.13496 preprint, 41 pages.
- Relatively non-degenerate integrated decay estimates for massless Vlasov fields on Schwarzschild spacetimes, with Léo Bigorgne; J. Éc. Polytech. Math 13 (2026). 82 pages.
- Late-time asymptotics of small data solutions for the Vlasov–Poisson system, with Léo Bigorgne; Nonlinearity 39 (2026), no. 4, 045007. 54 pages.
- A note on integrated local energy decay estimates for spherically symmetric black hole spacetimes, with Gustav Holzegel and Georgios Mavrogiannis; Math. Ann. 395, 44 (2026). 29 pages.

Modified scattering of small data solutions for the Vlasov–Poisson system with a repulsive potential, with Léo Bigorgne and Anibal Veloza Ruiz; SIAM J. Math. Anal. 57:1 (2025). 39 pages.

Small data solutions for the Vlasov–Poisson system with a repulsive potential, with Anibal Veloza Ruiz; Commun. Math. Phys. 405:80 (2024). 45 pages.

Gevrey regularity for the Vlasov–Poisson system; Ann. Inst. H. Poincaré Anal. Non Linéaire 38:4 (2021). 21 pages.

Characterization of uniform hyperbolicity for fibre-bunched cocycles; Dyn. Syst. 35:1 (2020). 16 pages.

CONFERENCE TALKS

September 2026 Conference “Geometric Analysis by the Andes”, Universidad de Chile and Pontificia Universidad Católica de Chile.

June 2026, Workshop “Long time dynamics of Vlasov equations”, Università degli Studi di Roma Tor Vergata.

January 2026, 2nd Münster–Imperial Day in PDE, Universität Münster.

January 2026, Segunda Jornada Anual de Nuevos Edepietas, Mathematical Engineering Department–Center for Mathematical Modeling, Universidad de Chile.

January 2026, Conference “Recent Progress in Asymptotic Stability of Solitons and Related Problems”, Universidad de Magallanes, Chile.

November 2024, Workshop in Dynamical Systems and Related Topics, Penn State University.

June 2023, MathInGP Postdoctoral Research Symposium, Sorbonne Université.

March 2023, MAFRAN Days, University of Cambridge.

December 2022, International Conference on Geometric Analysis and Hyperbolic Equations, Guangxi Center for Mathematical Research, Guangxi University.

November 2022, Workshop “Vlasov Equations: Geometric and Semi-classical Aspects”, Institut de Recherche Mathématique de Rennes.

April 2022, Oxbridge PDE Conference, University of Oxford.

December 2021, PDEs Session, Annual Meeting of the Chilean Mathematical Society, Universidad de O’Higgins.

December 2021, MAFRAN Days, Université Paris-Dauphine.

SEMINAR TALKS

October 2026, Department Seminar, Universidad Técnica Federico Santa María, Chile.

September 2026, Analysis & PDE Seminar, Università degli Studi di Roma Tor Vergata.

August 2026, PDE Seminar, Mathematical Engineering Department, Universidad de Chile.

July 2026, Calculus of Variations & PDE Seminar, Universidad de O’Higgins.

June 2026, Analysis and PDE Seminar, Gran Sasso Science Institute.

March 2026, Analysis Seminar, University of Birmingham.

March 2026, PDE and Mathematical Physics Seminar, ETH Zürich.

February 2026, Analysis Seminar, University of Bath.

February 2026, Analysis Seminar, EPFL.

November 2025, General Relativity & PDE seminar, Leipzig University.

October 2025, Applied PDE Seminar, Imperial College London.

June 2025, Calculus of Variations & PDE Seminar, Universidad de O'Higgins.

May 2025, Mathematical Physics Seminar, Pontificia Universidad Católica de Chile.

April 2025, Princeton Gravity Initiative Seminar Series, Princeton University.

April 2025, Geometric Analysis Colloquium, The Fields Institute.

July 2024, London PDE Seminar, Imperial College London.

June 2024, Analysis and Geometry Seminar, Pontificia Universidad Católica de Chile.

June 2024, PDE Seminar, Mathematical Engineering Department, Universidad de Chile.

March 2024, Analysis Seminar, Columbia University.

March 2024, Mathematical Physics Seminar, Johns Hopkins University.

November 2023, Princeton Gravity Initiative Seminar Series, Princeton University.

September 2023, Analysis and Applied Math Seminar, University of Toronto.

July 2023, Seminari di Metodi Analitici e Probabilistici, Università di Pisa.

April 2023, PDE Seminar, Mathematical Engineering Department, Universidad de Chile.

April 2023, Dynamical Systems Seminar of Santiago, Universidad Católica de Chile.

December 2022, Mathematical General Relativity Seminar, Sorbonne Université.

June 2022, London PDE Seminar, University College London.

May 2022, Topics in General Relativity, University of Münster.

April 2022, Early Career Math Colloquium, University of Arizona.

February 2022, PDE Seminar, Brown University.

January 2022, Analysis and Geometry Seminar, Pontificia Universidad Católica de Chile.

December 2021, Mathematical General Relativity Seminar, Sorbonne Université.

July 2021, Online Mathematical GR and Hyperbolic PDE Seminar.

June 2021, Junior Analysis Seminar, Imperial College London.

January 2021, Kinetic Theory Seminar, University of Cambridge.

November 2020, PDE Seminar, Mathematical Engineering Department, Universidad de Chile.

September 2019, Analysis and Geometry Seminar, Pontificia Universidad Católica de Chile.

July 2018, Dynamical Systems Seminar, Pontificia Universidad Católica de Chile.

July 2018, Dynamical Systems Seminar, Pontificia Universidad Católica de Valparaíso.

ACADEMIC VISITS

January 2027, Oberwolfach Workshop, “Wave-Kinetic Interfaces: Dispersive Dynamics Meets Kinetic Theory”, invited by Mikaela Iacobelli, Natasa Pavlovic, Gigliola Staffilani, and Nicola Visciglia (one week).

July 2025, Simons Center for Geometry and Physics, 3rd Simons Math Summer Workshop: “PDEs of Classical Physics”, invited by John Anderson, Mihalis Dafermos, Theodore Drivas, and Jonathan Luk (three weeks).

July 2024 and November 2024, Imperial College London, Department of Mathematics, invited by Martin Taylor (one week each time).

November 2019 and March 2022, Princeton University, Department of Mathematics, invited by Mihalis Dafermos (a month each time).

January 2022 and January 2025, Pontificia Universidad Católica de Chile, Department of Mathematics, invited by Mariel Sáez and Anibal Velozo Ruiz, respectively (a month each time).

TEACHING

Instructor, MA5612 Mathematical General Relativity, Universidad de Chile, Fall 2026.

Instructor, MA1002 Differential and Integral Calculus, Universidad de Chile, Fall 2026.

Instructor, MAT137 Calculus with Proofs, University of Toronto, Fall 2024, Winter 2025.

Instructor, MAT136 Calculus II, University of Toronto, Winter 2024.

Instructor, MAT244 ODEs, University of Toronto, Fall 2023.

Teaching Assistant, Part III Analysis & PDEs, University of Cambridge, Michaelmas 2021.

Supervisions, Part II Differential Geometry, University of Cambridge, Lent 2020, Lent 2022.

Supervisions, Part IB Electromagnetism, University of Cambridge, Lent 2021.

Supervisions, Part II Linear Analysis, University of Cambridge, Michaelmas 2020.

Supervisions, Part II Integrable Systems, University of Cambridge, Michaelmas 2020.

Teaching Assistant, Topology, Pontificia Universidad Católica de Chile, Spring 2017.

Teaching Assistant, ODEs, Pontificia Universidad Católica de Chile, Winter 2017.

Teaching Assistant, Multivariable Calculus, Pontificia Universidad Católica de Chile, Fall 2015.

Teaching Assistant, Calculus, Pontificia Universidad Católica de Chile, Spring 2014, Spring 2015.

ACADEMIC HONOURS

Scholarship PhD (2018–2022), CONICYT-Chile Cambridge Scholarship, Cambridge Trust.

Scholarship PhD (2018–2022), National Commission for Scientific and Technological Research, Chile.

Smith-Knight & Rayleigh-Knight Prize (2020), Faculty of Mathematics, University of Cambridge.

Scholarship MS (2018), National Commission for Scientific and Technological Research, Chile.

Scholarship MS (2017), Department of Mathematics, Pontificia Universidad Católica de Chile.

Teaching Assistant of First Category, Department of Mathematics, Pontificia Universidad Católica de Chile (2016).

Scholarship Rolando Chuaqui (2013–2016), Department of Mathematics, Pontificia Universidad Católica de Chile.

SERVICE

Referee of Articles for: Acta Applicandae Mathematicae, Acta Mathematica Vietnamica, Annales Henri Poincaré, Annals of PDE, Archive for Rational Mechanics and Analysis, Communications in Mathematical Physics, Communications in Partial Differential Equations, Journal of Differential Equations, Journal of Functional Analysis, Journal of Statistical Physics, SIAM Journal on Mathematical Analysis, Transactions of the American Mathematical Society.

Co-organiser PDE Seminar, Mathematical Engineering Department, Universidad de Chile (2026–).

Co-organiser London PDE Seminar, UK (2025–2026).

Co-organiser Paris Reading Group in General Relativity, Sorbonne Université (2022–2023).

Co-organiser MAFRAN Winter School: Particle Systems, PDEs and Inequalities, University of Cambridge (December 2020).

Co-organiser Cambridge Analysts' Knowledge Exchange Seminar, University of Cambridge (Lent and Easter terms 2020).

Organiser Differential Geometry Graduate Student Seminar, Pontificia Universidad Católica de Chile (2017–2018).

Co-author of 'Problems and Solutions in Single Variable Calculus', with Sebastian Burgos and Godofredo Iommi, 2026. 154 pages.

OUTREACH

Invited talk, Graduate School of Churchill College, University of Cambridge (May 2021).

Instructor, Course "Introduction to Mathematical Reasoning" (TRM), special program for high school students. Pontificia Universidad Católica de Chile (2017–2018).

"da Vinci Dome", Stand at Math Festival, San Antonio, Chile (September 2017).

Instructor, Chilean Team for International Math Olympiads IMO and Iberoamerican (2016–2017).

Tutor, Program "Fun Math", National Service for Minors of San Joaquín, Pontificia Universidad Católica de Chile (2017).

PERSONAL INFORMATION

Born: 16 March 1995, Santiago, Chile.

Citizenship: Chilean.

Languages: Spanish (native), English (fluent).